

ABHISHEK SEN

ROBOTICS ENGINEER

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PROFESSIONAL SUMMARY

Robotics Engineer specializing in autonomous navigation, ROS2 stacks, embedded hardware R&D, and Edge AI. Proven experience designing and deploying GPS-less drone guidance systems, sensor fusion (LiDAR, IMU, Camera), and path planning algorithms. Experienced Captain of Aerobotics Team Astra with accolades from ISRO and SAE India.

EDUCATION

B.Tech in Electronics & Communication

IIITDM Kancheepuram | Batch: 2022 - 2026

CGPA: 8.45 / 10.00 | Roll: EC22B1098

TECHNICAL SKILLS

Robotics & Navigation: ROS2 Humble, MAVROS, SLAM, GMapping, Gazebo, RViz2, ArduPilot

AI / ML / Vision: YOLOv8, PyTorch, TensorFlow, OpenCV, Edge AI (Jetson Orin Nano, TDA4VM)

Languages & Tools: Python, C/C++, Docker, Git, Altium, PCB Design

PROFESSIONAL EXPERIENCE

Robotics Engineer

Jun 2026 – Present

RobotSpace, Hyderabad

- Designing and developing robust autonomous mobile robot (AMR) platforms and navigation stacks using ROS2.
- Integrating state-of-the-art sensor suites, actuators, and Edge AI modules for real-world industrial deployments.

Project Assistant — Guidance, Navigation & Control Lab

Oct 2025 – May 2026

IIT Madras

- Implemented a proprietary path-planning system using the **Generalized Shape Expansion (GSE)** algorithm, showing significant speed and accuracy improvements over traditional RRT*.
- Deployed navigation stack on NVIDIA Jetson Orin Nano using a Livox 4D LiDAR for real-time localization and mapping.
- Developed sensor fusion filters (Kalman & Particle Filters) to combine IMU, wheel odometry, and LiDAR inputs.

Hardware R&D; Intern

May – Sep 2025

32Mins Digital Consultancy Services (IIT Madras Research Park)

- Designed an AI-driven classroom engagement device; developed dual-camera calibration systems on Texas Instruments TDA4VM board.
- Authored customized camera drivers deployed using Docker containers for Edge AI pipelines.

Project Intern

May – Jul 2024

Autonomous Systems Laboratory

- Developed autonomous ground vehicles implementing the GMapping algorithm for high-fidelity 2D grid mapping.

PORTFOLIO PROJECTS

- **Autonomous Docking & Visual Servoing Robot**

Developed a ROS2 package executing camera-guided autonomous docking in Gazebo. Fuses camera object tracking with LiDAR LaserScan distance reading inside a custom proportional control loop to enable high-precision alignment and docking.

- **Robust Navigation & Trajectory Control**

Created a custom lightweight navigation stack for TurtleBot3 using B-Spline path smoothing to maintain C2 trajectory continuity. Integrated a Pure Pursuit path tracking controller operating at 20Hz for high-speed obstacle avoidance.

- **Autonomous Drone — Martian Terrain Analysis (ISRO Challenge)**

Built a GPS-less autonomous drone platform using Intel RealSense Visual SLAM for real-time 3D localization. Implemented YOLOv8 Edge AI on-board inference for Mars analogue terrain/boundary classification and obstacle avoidance.

- **Autonomous Surveillance & Object Tracking Drone**

Designed a custom quadcopter UAV hardware chassis, power distribution, and software stack from scratch. Integrated YOLOv8 deep learning models for autonomous search, tracking, and target lock-on operations.

KEY ACHIEVEMENTS & COMPETITIONS

ISRO Robotics Challenge (IRoC-U 2025): Selected as one of the **Top National Finalists**. Earned personal appreciation from **ISRO Chairman Dr. S. Somnath** at Bharat Mandapam, New Delhi, and demonstrated the autonomous Martian drone to Indian Astronaut Angad Pratap.

SAE Aerothon (2024 & 2025): Secured **All India Rank 8** (AIR 8) in Detailed Design Report & Presentation Round in 2025. Won the **National Challenger Round Championship** in 2024, and ranked AIR 21 in report documentation.

IRoC-U 2025 Final Field Round (IIT Roorkee): Represented IIITDM Kancheepuram as one of only 4 national teams to successfully complete both manual piloting and fully autonomous flight missions in outdoor Martian analogue environments.

LEADERSHIP & POSITION OF RESPONSIBILITY

Team Captain — Aerobotics Team Astra (IIITDM Kancheepuram) | Jul 2023 – Jan 2026

Led a cross-disciplinary team of 25+ engineers. Guided hardware R&D, firmware development, autonomous ROS2 software architecture, and budget mapping. Spearheaded institution representation at top-tier space tech festivals.

Active NCC Cadet — National Cadet Corps | 2023 – 2025

Honed leadership, discipline, and community development frameworks through rigorous training camps.